



Assess and Guide: Multi-modal Fake News Detection via Decision Uncertainty

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Abstract

The prevalence of fake news, in both visual and textual forms, has become a critical and urgent issue. Existing multi-modal fake news detection methods aim to improve detection performance by leveraging information from different modalities and integrating it into a unified, comprehensive representation. However, challenges arise when different modalities hold conflicting beliefs. This suggests the existence of information asymmetry and imbalance between various modalities, which has not been adequately recognized in previous work. In this regard, focusing on the uncertainty of each modality facilitates a more nuanced understanding of its reliability and informativeness. In this work, we distinguish among uni-modal information sources through accurate estimation of their decision uncertainty, employing this uncertainty as a confidence measure and indicator of information imbalance to guide feature fusion. In particular, 1) To provide accurate uncertainty estimation, an

uncertainty learning method based on the Dirichlet distribution is employed; 2) To fuse multi-modal information, uncertainty is used as a guiding factor to govern the contribution of each modality. Extensive experiments on two real-world datasets WEIBO and TWITTER validate the effectiveness of the proposed method.

CCS Concepts

• **Applied computing** → *Sociology*; • **Information systems** → **Social networks**; • **Computing methodologies** → **Information extraction**.

Keywords

Multi-modal Learning, Fake News Detection, Uncertainty Learning

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1 Introduction

Fake news refers to information that is deliberately fabricated to deceive people [11]. In recent years, the detection of fake news has become an increasingly critical topic, as misinformation spreads

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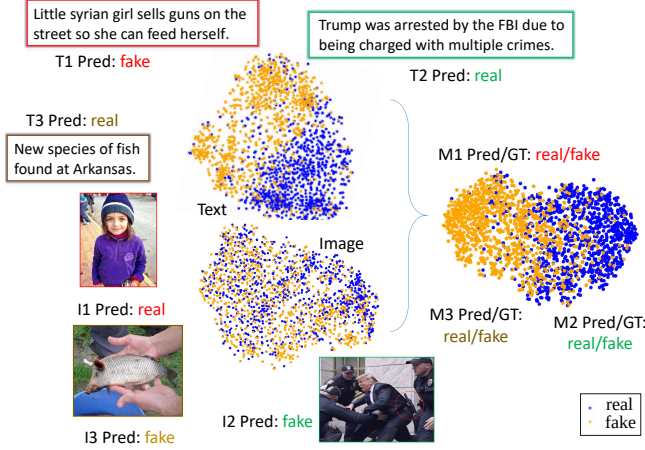


Figure 1: T-SNE visualization of uni-modal and multi-modal mined features on WEIBO test set. Text modality exhibits stronger discriminative power, while images appear somewhat ambiguous. This imbalance often leads to inconsistent predictions, ultimately resulting in incorrect classification.

more rapidly and widely than the truth on social media. The widespread dissemination of fake news poses significant threats to individuals and society alike. However, accurately identifying fake news remains a significant challenge [32]. Given this pressing scenario, the automated detection of fake news has become imperative for the sustainable and healthy development of news platforms.

In online news platforms such as Weibo and Twitter, daily published posts contain a variety of modalities, including images, text, videos, and more. From a uni-modal perspective, there are machine learning methods, such as using decision trees [20], as well as neural network approaches like analyzing writing style [23], utilizing attention mechanisms [2], and extracting features from the frequency domain [24]. However, uni-modal methods are somewhat disadvantaged in dealing with the increasing prevalence of multi-modal fake news, since they ignore correlations among different modalities, resulting in insufficient utilization of multi-modal information. To address this deficiency, researchers have turned their focus to the field of multi-modal fake news detection.

To explore the semantic consistency between images and text, existing multi-modal methods typically reduce the distance between image-text features using similarity-aware techniques and map them into a shared latent space, aiming to obtain aligned and unified feature representations for detection [3, 8, 37]. Semantic consistency has proven helpful in fake news detection, and methods based on this perspective have yielded promising results [1, 15, 18, 27].

However, while focusing on semantic consistency and treating uni-modal information equally, they often overlook the common information imbalance between text and images. In particular: 1) while the text modality typically exhibits better discriminative power, the image modality generally suffers from semantic ambiguity and greater noise, resulting in an indistinguishable and uncertain T-SNE distribution, as shown in Fig 1; 2) as illustrated by the three examples in Fig 1, it is common for images and text



Figure 2: Different modalities of the same news exhibit varying levels of uncertainty. The fusion process choose to utilize more information from the modality with lower uncertainty score and make a joint decision.

to make opposite predictions, leading to incorrect classification results after the fusion process.

Inspired by the prevalent issue of imbalanced information in multi-modal fake news data, both text and image modalities require individualized processing based on their unique characteristics and call for authoritative guidance in the subsequent integration process. To address this challenge, we accurately estimate the decision uncertainty of each uni-modal information source, employing this uncertainty as a confidence measure and indicator of information imbalance to guide subsequent feature fusion. As shown in Fig 2, different modalities within the same news pair exhibit varying uncertainty levels. Specifically, the red box in the image highlights deceptive information (e.g., a president paired with a gun), while the text mistakenly labels the content as "real" due to the lack of such obvious misinformation. This discrepancy indicates a lower decision uncertainty in the image modality, making it more likely to produce confident predictive outcomes. If we were to perform a simple average fusion, the result might be biased by the text and lead to a decision contrary to the ground truth. However, by incorporating uncertainty assessment, the image modality is assigned a lower uncertainty score, meaning its features will be given more weight during the subsequent fusion stage.

Building on this concept, we present a new framework called T-FND. T-FND begins by using the outputs of uni-modal detectors to model the parameters of a Dirichlet distribution, allowing for accurate uncertainty estimation for each modality. Unlike previous fusion approaches, T-FND employs these uncertainty estimates to guide the fusion of uni-modal features. In this process, modalities with higher uncertainty are down-weighted, while those with lower uncertainty are given greater emphasis. By accurately assessing uncertainty and employing uncertainty-based feature fusion, T-FND effectively addresses modal imbalances, thereby enhancing the reliability and accuracy of multi-modal fake news detection.

In summary, the specific contributions of this paper are:

- We observe that information imbalance is prevalent in multi-modal fake news data. To address this issue, we propose a new framework called T-FND, which measures and leverages

this imbalance through uncertainty estimation and uses this uncertainty to guide the fusion process.

- We introduce a new multi-modal fusion approach in which decision uncertainty guides the fusion of multi-modal features. Compared to mainstream methods, our approach demonstrates improved performance in fake news detection.
- Extensive experiments on two widely used real-world datasets, TWITTER and WEIBO, demonstrate the effectiveness of the proposed approach.

2 Related work

2.1 Fake News Detection

Fake news detection is defined as the process of using AI to determine the authenticity of a given social media post and categorize it as either real or fake. Existing fake news detection methods can be broadly categorized into two approaches: uni-modal and multi-modal. With the growth of social networks, there is a trend for news to shift from uni-modal to multi-modal formats. Multi-modal fake news poses greater challenges.

2.1.1 Uni-modal Approach. Both language-based and vision-based fake news detection have received extensive attention. The study of text misinformation has undergone significant evolution, transitioning from traditional machine learning methods to more advanced deep learning models and their combinations [6]. In recent years, there has been a notable shift towards the adoption of transformers and pre-trained language models like BERT [4] for misinformation detection, resulting in improved performance outcomes. Apart from deepfake [21] studies, the sole visual modality has received relatively less attention. A recent example is the work of MVNN [25], which innovatively integrates spatial and frequency domain features of news for forensic analysis. However, for the multi-modal news, these approaches are unable to detect cross-modal correlations.

2.1.2 Multi-modal Approach. Recently, there have been notable advancements in fake news detection methods that leverage cross-modal discriminative patterns to achieve superior performance. One such method is EANN [29], which incorporates an additional event discriminator to aid in feature extraction and learn cross-modal characteristics. Another approach, MVAE [15], introduces a multi-modal variational autoencoder that learns probabilistic latent variable models and reconstructs the original texts and low-level image features. Similarly, MCAN [31] utilizes multiple co-attention layers to effectively fuse textual and visual features for accurate fake news detection. CMC [30] uses a two-stage network that trains two uni-modal networks to learn cross-modal correlations by contrastive learning, and then fine-tunes the network for fake news detection.

Additionally, there are methods that utilize both uni-modal and multi-modal information. For instance, CAFE [3] measures cross-modal ambiguity by evaluating the Kullback-Leibler (KL) [16] divergence between the distributions of uni-modal features. The learned ambiguity score is then used to linearly adjust the weights of uni-modal and multi-modal features before final classification. However, CAFE assigns equal weights to each modality, whereas the confidence levels of different modalities should not be the same. Another

method, LIIMR [31], identifies the modality that exhibits greater confidence in fake news detection.

2.1.3 Uncertainty-based Learning. Uncertainty quantification (UQ) has become an increasingly important component in developing trustworthy machine learning systems. It has been successfully applied across various downstream tasks [9, 10, 17, 19, 26]. Accurate uncertainty estimation is essential for improving the reliability of predictive models, particularly in high-stakes applications such as medical diagnosis and, potentially, multi-modal fake news detection. Recently, evidential deep learning has been introduced as a promising approach, using a single neural network and a single forward pass to estimate prediction uncertainty at a given point. Uncertainty receives attention as it can serve as a measure of confidence in each modality and provide a qualitative indicator of information imbalance. In this work, to address the issues of imbalance in multi-modal data and achieve reliable classification, we leverage uncertainty at the decision level as guidance for fusing multi-modal features. When one modality lacks confidence in identifying misinformation, its respective view is down-weighted in the subsequent fusion process.

3 Proposed Approach

The overview of the proposed T-FND is shown in Fig 3. We design a dual-stream network: one branch is dedicated to accurately estimating the uncertainty of each modality, while the other focuses on feature fusion based on this uncertainty to provide the final classification result. In the uncertainty estimation module, we use a Dirichlet distribution-based uncertainty learning method, where higher uncertainty reflects lower confidence in the corresponding modality. In the feature fusion and decision module, we incorporate each modality's uncertainty as a linear guiding factor for fusion, with modalities of higher uncertainty being down-weighted and those of lower uncertainty receiving greater emphasis. By considering uncertainty, we can effectively weight each modality's contribution during the fusion process, resulting in more accurate and reliable outcomes.

For each piece of news x , we leverage a pretrained Vision Transformer [7] and BERT [5] to encode the image and text into uni-modal embeddings x^v and x^t , respectively.

3.1 Uncertainty Estimating Module

In this section, we will discuss how to perform accurate uncertainty estimation. After obtaining the image features x^v and text features x^t , we pass them through two separate uni-modal classifiers to obtain the classification probabilities for each modality. The two uni-modal classifiers output their respective pieces of predictive evidence $e^t = [e_r^t, e_f^t]$ (where e_r^t means the probability of classifying a sample as "real" in the text modality) and $e^v = [e_r^v, e_f^v]$.

Subjective logic [14] associates the parameters of the Dirichlet distribution with the predicted probabilities. Accordingly, we use $\alpha^t = [e_r^t + 1, e_f^t + 1]$ as the construction parameter for the Dirichlet distribution of the text modality. The belief mass and the uncertainty scores are computed as follows:

$$b_r^t = \frac{e_r^t}{S^t}, \quad b_f^t = \frac{e_f^t}{S^t}, \quad u^t = \frac{K}{S^t}, \quad (1)$$

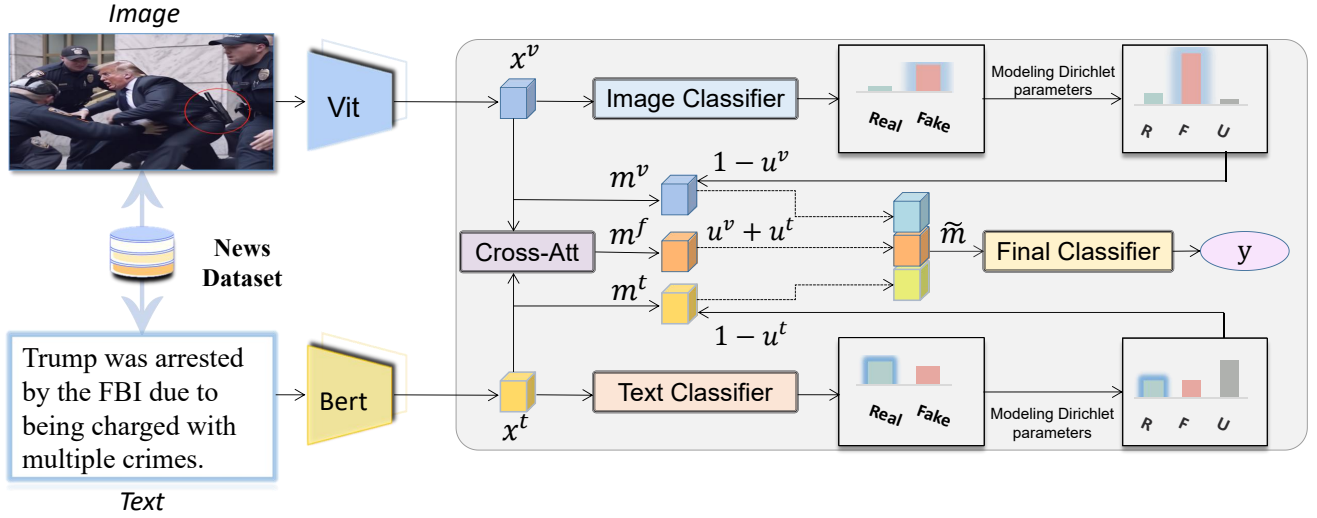


Figure 3: The architecture of the proposed T-FND. The model consists of an uncertainty estimation branch (upper and lower parts) and a multi-modal feature fusion branch (the middle part). In the uncertainty estimation branch, text embedding x^t and image embedding x^v are individually passed through uni-modal classifiers to generate predictions. These predictions are then used to parameterize a Dirichlet distribution, deriving the uncertainty scores u^t and u^v . The feature fusion module utilizes uncertainty scores to guide the multi-modal fusion process, incorporating projected uni-modal features m^t , m^v and fused representation m^f obtained via a cross-attention mechanism. Concatenated feature $\tilde{m}$ is fed into the final classifier.

where u^t and b_i^t indicate the overall uncertainty of text modality and the probability for the "real" classification, respectively. $S^t = \alpha_f^t + \alpha_r^t$ is the Dirichlet strength. K represents the number of categories, which is two in our case. Similarly, for the image modality:

$$b_r^v = \frac{e_r^v}{S^v}, \quad b_f^v = \frac{e_f^v}{S^v}, \quad u^v = \frac{K}{S^v}, \quad (2)$$

Next, we discuss how to train neural networks to achieve accurate uncertainty calibration. Traditional cross-entropy loss is commonly used for classification tasks:

$$\mathcal{L}_{ce} = - \sum_{i=1}^K y_i \log(p_i), \quad (3)$$

where p_i is the predicted probability for class i .

Inspired by [9], we made a minor modification to the cross-entropy loss by multiplying it with the probability density of the Dirichlet distribution:

$$\mathcal{L}_{ce}(\alpha^t) = \int \left(\sum_{i=1}^K -y_i \log(b_i^t) \right) \frac{1}{B(\alpha^t)} \prod_{i=1}^K (b_i^t)^{\alpha_i^t - 1} db^t,$$

i can be either real or fake.

The uncertainty calibration loss for the text modality is:

$$\mathcal{L}(\alpha^t) = \mathcal{L}_{ce}(\alpha^t) + \lambda \text{KL} [D(b^t | \tilde{\alpha}^t) || D(b^t | 1)], \quad (4)$$

where $\tilde{\alpha}_i = y_i + (1 - y_i) \odot \alpha$ is the adjusted parameter of the Dirichlet distribution that avoids penalizing the probabilities of the ground truth class to 0. λ factor gradually increases from 0 to 1 as the training progresses. When λ is small, it allows for more flexible exploration of the parameter space, accelerating convergence; as λ

increases, the focus shifts to more accurate uncertainty estimation. The total uncertainty estimation loss is as follows:

$$\mathcal{L}_{ue} = \sum_{j=1}^N \left[\mathcal{L}(\alpha_j^v) + \mathcal{L}(\alpha_j^t) \right], \quad (5)$$

where N is the number of samples in a batch. The uncertainty assessment provided by this module will provide valuable guidance for our following feature fusion.

3.2 Multi-Modal Feature Fusion Module

To achieve better feature fusion, we need to explore deep-level feature information. In this module, we project the features x^t and x^v into a lower-dimensional space to uncover deep-level feature information for fusion. To facilitate subsequent fusion, both the textual and visual modalities are projected into the same dimensional space. The projection structure is a multi-layer perceptron (MLP), consisting of Linear–BatchNorm–ReLU–Linear, with dimensions from $\mathbb{R}^{768}$ to $\mathbb{R}^{384}$ and then to $\mathbb{R}^{64}$. This allows for efficient integration and fusion of the two modalities. The uni-modal embeddings are denoted as m^v for the image modality and m^t for the text modality after projection.

We perform cross-attention [28] between m^v and m^t to obtain fused embeddings that capture image-text correlations. The attention function is computed on the normalized query (Q), key (K), and value (V) features as follows:

$$\text{Attention}(Q, K, V) = \text{Softmax} \left(\frac{K^T Q}{\sqrt{D}} \right) V, \quad (6)$$

where D is 64. Here, we cross-attend m^v with the text embedding m^t by treating the image embedding as the query (Q) and the text embedding as both the key (K) and value (V) to obtain image embeddings enriched with textual information. Similarly, we perform cross-attention to obtain a text embedding that incorporates image information. These fused embeddings are then concatenated to form the multi-modal supplementary information m^f .

At this stage, we have the confidence scores u^v and u^t provided by the uncertainty estimation module, the uni-modal features m^v and m^t from the two modalities, and the feature m^f representing the cross-modal interaction.

Traditional fusion methods, whether through simple concatenation or attention-based mechanisms, tend to treat different modalities equally, without accounting for the uncertainty inherent in each uni-modal representation. CAFE [3] quantifies cross-modal ambiguity by assessing the KL divergence between the distributions of uni-modal features. The resulting ambiguity score is then used to linearly adjust the weights of uni-modal and multi-modal features prior to final classification. However, the modality fusion used by CAFE imposes a constraint that the text and image modalities share the same weight. When the uncertainty of a modality is high, we should not rely heavily on information from that modality to make decisions. Conversely, when the confidences of both modalities are low, it is necessary to integrate both modalities to make a comprehensive judgment. Based on this, we propose a fusion method driven by uncertainty.

In T-FND, the input to the final classifier is obtained by adaptively concatenating two sets of embeddings: one set consists of the uni-modal representations u^v and u^t , and the other set consists of the cross-modality correlations m^f from the cross-attention fusion module. This concatenation process is governed by the uncertainty scores u^v and u^t provided by the aforementioned uncertainty evaluation module, and it is described as follows:

$$\tilde{m} = ((1 - u^v) \times m^v) \oplus ((1 - u^t) \times m^t) \oplus ((u^v + u^t) \times m^f), \quad (7)$$

where $\oplus$ represents the concatenation operation. The fusion method ensures: 1. when the uncertainty of a uni-modal representation is high, we rely less on information from that modality. 2. when the uncertainties of both modalities are high, the uni-modal information is scarcely used, while the cross-modal interaction information m^f is given more prominence. Based on this, we aim to ensure that the features fed into the classifier can automatically adapt to individual samples and provide reliable classification.

Then, we feed the final representation $\tilde{m}$ into a Multi-Layer Perception (MLP) to predict the label as $\tilde{y}$:

$$\tilde{y} = \text{softmax}(\text{MLP}(\tilde{m})) \quad (8)$$

The loss for classification is commonly calculated using cross-entropy with the ground truth label y :

$$\mathcal{L}_{cls} = y \log(\tilde{y}) + (1 - y) \log(1 - \tilde{y}). \quad (9)$$

In order to simultaneously provide accurate uncertainty assessment and effective classification, we employ multi-task training strategy. The final loss function is given by:

$$\mathcal{L} \triangleq \mathcal{L}_{uc} + \mathcal{L}_{cls}. \quad (10)$$

4 EXPERIMENTS

This section presents the implementation issues and experimental results of applying our method to real world datasets and its comparison with advanced methods.

4.1 Datasets

We utilized two publicly available datasets collected from social media, as described below.

4.1.1 WEIBO. This dataset was collected from WEIBO social media from 2012 to 2016 and is written in Chinese. Each line of data contains text, and an image. The training set contains 7,532 news, including 3,749 fake news and 3,783 non-fake news while the test set contains 1,996 posts. So some methods such as EANN [29] aim to improve the quality of the dataset by filtering out challenging samples, including screenshots, images smaller than 64x64 pixels, and text with fewer than 5 words.

4.1.2 TWITTER. This dataset was introduced in for fake news detection on TWITTER. Data consists of a training set and a test set. Each news data has text, and some news items are accompanied by images. The training set contains 6,649 real news and 4,599 fake news and the test set contains 989 posts.

4.2 Baseline Method

We compared our method with both uni-modal and multi-modal methods for fake news detection. Here are the basic details of baseline used in our experiments.

- **U1.CAR** [2] combines recurrent neural networks (RNN) with an attention mechanism to capture relatively important textual information for detecting text-only fake news.
- **U2.VS** [13] explores the role of image content in automatic news verification on microblogs, proposing visual and statistical features to detect fake news based on distinct image distribution patterns between real and fake news events
- **M1.RA** [12] proposes an end-to-end att-RNN to effectively detect rumors by fusing multi-modal features, including text, image, and social context.
- **M2.EANN** [29] addresses the challenge of detecting fake news on newly emerged events by deriving event-invariant features through a multi-modal feature extractor, fake news detector, and event discriminator.
- **M3.SAFE** [36] which uses a pre-trained image-to-text model to transform the image into text, and then measures the similarity to detect fake news.
- **M4.MKEMN** [35] fuses text, images, and retrieved information using CNN.
- **M5.MCNN** [33] uses cosine similarity to compute text-image similarity to assist in fake news detection.
- **M6.MVAE** [15] uses a variational autoencoder with a binary classifier to model representations between text and images.
- **M7.CAFE** [3] takes into account the KL divergence of the distributions of image and text modalities to guide modality fusion.

Table 1: Performance comparison between ours and baseline methods. Bold face indicates the best overall performance.

	Method	Acc	Fake news			Real news		
			Pre	Rec	F1	Pre	Rec	F1
TWITTER	CAR	0.637	0.574	0.690	0.682	0.724	0.602	0.617
	VS	0.617	0.635	0.644	0.639	0.639	0.630	0.634
	RA	0.664	0.749	0.615	0.676	0.589	0.728	0.651
	EANN	0.648	0.810	0.498	0.617	0.584	0.759	0.660
	MVAE	0.745	0.801	0.719	0.758	0.689	0.777	0.730
	MKEMN	0.715	0.814	0.756	0.708	0.634	0.774	0.660
	SAFE	0.762	0.831	0.724	0.774	0.695	0.811	0.748
	MCNN	0.784	0.778	0.781	0.779	0.790	0.787	0.788
	CAFE	0.806	0.807	0.799	0.803	0.805	0.813	0.809
	Ours (T-FND)	0.824	0.772	0.918	0.838	0.899	0.730	0.806
WEIBO	CAR	0.745	0.705	0.765	0.750	0.756	0.725	0.740
	VS	0.726	0.732	0.712	0.722	0.720	0.740	0.730
	RA	0.772	0.854	0.656	0.742	0.720	0.889	0.795
	EANN	0.795	0.806	0.795	0.800	0.752	0.793	0.804
	MVAE	0.824	0.854	0.769	0.809	0.802	0.875	0.837
	MKEMN	0.814	0.823	0.799	0.812	0.723	0.819	0.798
	SAFE	0.816	0.818	0.815	0.817	0.816	0.818	0.817
	MCNN	0.823	0.858	0.801	0.828	0.787	0.848	0.816
	CAFE	0.840	0.855	0.830	0.842	0.825	0.851	0.837
	Ours (T-FND)	0.888	0.900	0.872	0.886	0.877	0.904	0.890

4.3 Implementation Details

4.3.1 Experimental Setups. All our experiments were conducted on a single NVIDIA V100 GPU using the PyTorch [22] framework. The image encoder utilized ViT-B/16, while the text encoder employed BERT-base. The two uni-modal classifiers in the uncertainty estimation module were implemented as multi-layer perceptron (MLP) layers with a softplus function as the output activation function. The classifier in the feature fusion module used a fully connected layer with a softmax activation function. Training employed an early stopping strategy to prevent overfitting, and learning rate decay was managed using ReduceLROnPlateau. The experiments were conducted with a batch size of 32, and AdamW was used as the optimizer for training over 50 epochs. We used the same split ratio (7:1:2 for training, validation, and testing) for WEIBO, following EANN [29]. For TWITTER, an eighth of both real and fake training samples form the validation set. Similar to EANN [29] and BMR [34], text that is either absent or shorter than five words is replaced with "No text provided", while images smaller than 64×64 pixels are padded with a *zero matrix*.

4.3.2 Evaluation Metric. The evaluation metrics we used are consistent with those in previous multi-modal fake news detection works: accuracy, recall, precision, and F1-score.

4.4 Comparison with baselines

Table 1 presents the comparison results on WEIBO and TWITTER with various baselines. T-FND achieves the highest accuracy and F1 score across both real-world datasets, with particularly strong performance on the WEIBO dataset.

Specifically, Table 1 allows us to draw the following conclusions: Multi-modal methods generally outperform text-only or image-only

methods across various metrics, highlighting the benefits of fusing and cross-referencing different modalities for fake news detection. Among various multi-modal methods, CAFE [3] performs the best as it addresses the information gap between text and images by modeling this gap with the distance between Gaussian distributions derived from text and image features. We observed increases of 4.6% and 1.8% in accuracy on the WEIBO and TWITTER datasets, respectively, demonstrating the effectiveness of the proposed T-FND method.

It can be observed that the gains of our method vary across different datasets, with a much stronger performance advantage on the WEIBO dataset compared to TWITTER. We infer this might be because, in the WEIBO dataset, images and text are well-paired, which facilitates our framework in conducting a comprehensive uncertainty assessment based on both modalities. However, for the TWITTER dataset, the number of images is significantly smaller compared to the volume of text data, causing our method to degrade into a uni-modal approach and its performance gains cannot be demonstrated as effectively as in WEIBO. It is worth noting that other methods also face the same challenge, and generally, performance on the TWITTER dataset tends to be slightly lower than on the WEIBO dataset.

5 Ablation Study

5.1 Ablation of uncertainty estimating module

To validate the effectiveness of the uncertainty learning module, we sequentially removed uncertainty learning for each modality and eliminated its contribution to modality fusion as follows:

- w/o u^v . We removed the uncertainty learning for the image modality and set the weight of the image modality to 1.0

during the fusion of multi-modal information, instead of using its original weight ($1.0 - u^v$).

- w/o u^t . Similarly, we removed the uncertainty learning for the text modality and adjusted the weight during modality fusion.
- one branch. We removed all uncertainty-related module, using directly concatenated uni-modal features as multi-modal feature for detection.

From Table 2, we observed that removing the uncertainty estimation for a particular modality results in a decrease in accuracy by 2% to 2.2%, demonstrating that the uncertainty of each modality plays a role in aiding multi-modal fusion. Even after removing the uncertainty estimation for both modalities, we still achieve an accuracy of 85.9% thanks to the powerful capabilities of the pre-trained transformer.

Table 2: Ablation study on the uni-modal uncertainty estimating module on the WEIBO datasets.

Method	Acc	Pre	Rec	F1
w/o u^v	0.868	0.869	0.868	0.868
w/o u^t	0.864	0.867	0.865	0.864
one branch	0.859	0.862	0.859	0.859
Ours	0.888	0.888	0.888	0.888

5.2 Ablation study of multi-modal fusion

To validate the rationale for using uncertainty as a basis for modality fusion, we evaluated the performance of the model under different fusion approaches. We tested three fusion methods: (1) assigning fixed weights to text and image modalities, (2) allocating weights based on cross-modal attention weights, and (3) our proposed approach of weight allocation based on uncertainty. Additionally, we included ablation experiments for specific modalities to provide a comprehensive comparison.

Specifically, our experimental design is as follows:

- On the left of Fig 4, we compared the fusion method using fixed weights.
- On the right side of Fig 4, we performed modality ablation and compared the effects of fusion based on uncertainty and fusion based on attention.

From Table 3, we observed interesting results. The modality ablation experiments confirm a significant imbalance among different modalities. In the WEIBO dataset, the textual modality provides clearer and more useful information compared to the visual modality. In the TWITTER dataset, the significantly smaller number of images relative to text also leads to an imbalance in modality information.

Fusing using a fixed ratio performs better than using uni-modal information alone, indicating that multi-modal fusion captures shared information across modalities and yields good results. However, a fixed ratio is not suitable for all samples, as it lacks adaptability to individual samples and reduces generalization ability. Fusion based on cross-attention weights did not perform as well as expected, likely due to a significant semantic gap between the two

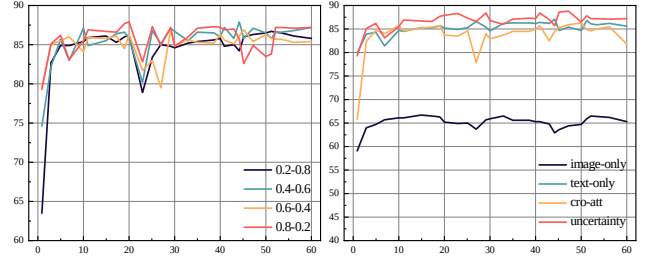


Figure 4: Curves of testing accuracy (y-axis) versus training epochs (x-axis). We examined the effects of assigning fixed weights to different modalities (left), modality ablation, and comparisons with cross-attention fusion (right).

Table 3: Ablation study of different multi-modal fusion methods on WEIBO datasets.

w_t	w_v	Acc	Pre	Rec	F1
0.2	0.8	0.865	0.867	0.865	0.864
0.4	0.6	0.878	0.878	0.879	0.878
0.6	0.4	0.869	0.870	0.869	0.869
0.8	0.2	0.872	0.874	0.872	0.872
1.0	0.0	0.856	0.859	0.857	0.856
0.0	1.0	0.667	0.666	0.667	0.667
att_{v2t}	att_{t2v}	0.863	0.864	0.863	0.863
$1 - u_t$	$1 - u_v$	0.888	0.888	0.888	0.888

modalities, which hinders the attention mechanism from accurately capturing useful information for detection.

Our fusion method based on uncertainty outperforms the aforementioned approaches. By performing adaptive uncertainty evaluation on each modality for every sample, it uses the uncertainty of each modality to guide the fusion process, achieving reliable and effective results.

6 Conclusions

In this paper, we introduce a new framework called T-FND to address the issue of information imbalance in multi-modal fake news detection. Unlike previous methods that primarily focus on semantic consistency, T-FND leverages uncertainty scores derived from a Dirichlet distribution to assess the confidence level of each modality. These uncertainties are then used to guide the fusion process of uni-modal features, ensuring more reliable and accurate classification outcomes. Experimental results on two widely used datasets, TWITTER and WEIBO, highlight the importance of considering modality-specific uncertainty to improve multi-modal fake news detection.

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